



Nutritional Analysis and Strategic Menu Engineering for McDonald's India

BDM Capstone Viva Presentation —
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A concise, data-driven audit and strategic roadmap to improve menu nutrition while preserving commercial performance.

DOMAIN: B2C

About McDonald's India



Leading QSR Network

Over 800 outlets across 75+ cities with strong brand recognition and scale.



Glocal Strategy

Localized menu adaptations combined with global operating standards.



Operational Strength

Robust supply chain and franchise operations enabling rapid execution.



Customer Focus

Continuous emphasis on experience, value, and menu innovation.

This slide sets context for why menu-level nutritional optimization is both operationally feasible and strategically important.

Business Problem and Objectives

Business Challenge

The Health–Profit Paradox: reconcile popular high-margin items with rising demand for healthier choices.

Objective 1

Identify nutritionally inefficient menu items for targeted pruning.

Objective 2

Design healthy, commercially viable meal bundles (Nutri-Bundles).

Objective 3

Measure satiety efficiency and build a Nutrient Density Index (NDI) for tiered labelling.

The analytics program balances nutritional improvement with business impact — focusing on feasibility, customer acceptability, and profitability.

Dataset and Analytical Framework

Dataset

- Source: Kaggle — McDonald's India menu nutrition dataset
- Scope: 141 menu items; fields include Energy, Protein, Sodium, Serving Size
- Type: Secondary, cross-sectional nutrition records

Analytical Workflow

1. Data collection and cleaning
2. Descriptive statistics & Z-score outlier detection
3. Correlation and satiety efficiency analysis
4. Bundle optimization and NDI development

Data Cleaning and Standardization

The screenshot shows a spreadsheet with columns for Menu Category, Menu Items, Per Serve Size, Energy (kCal), Protein (g), Protein-to-Weight Ratio, Total fat (g), Sat Fat (g), and Trans fat (g). The data is organized into rows for different menu categories like Beverages, Breakfast, McCafe, Regular, and Condiments. A 'Find and Replace' dialog box is open, showing the 'Replace' tab with the search term 'Find' and the replacement text 'Insert Text'.

Menu Category	Menu Items	Per Serve Size	Energy (kCal)	Protein (g)	Protein-to-Weight Ratio	Total fat (g)	Sat Fat (g)	Trans fat (g)
Beverages Menu	Vedica Natural Mineral Water	500.00	0	0	0	0	0	0
Breakfast Menu	Sausage Mc Muffin with egg	157.00	290.41	11.49				
McCafe Menu	Strawberry Green Tea (L)	456.00	8.32	7.03				
McCafe Menu	Strawberry Green Tea (R)	330.00	248.76	51.03				
McCafe Menu	Strawberry Green Tea (S)	279.00	443.4	164.44				
Regular Menu	McSpicy Fried Chicken 1 pc	115.00	58.00	9.93				
Condiments Menu	Cheese Slice	14.00	11.75	16.23				
Gourmet Menu	Piri piri Mc Spicy Chicken Burger	228.00	64.00	246.65				
Regular Menu	2 piece Chicken Strips	58.00	96.00	330.00				
McCafe Menu	English Breakfast (S)	279.00	7.39	10.21				
McCafe Menu	English Breakfast (R)	330.00	381.77	281.44				
Regular Menu	4 piece Chicken McNuggets	64.00	16.23	246.65				
McCafe Menu	English Breakfast (L)	456.00	87.00	96.00				
Regular Menu	3 piece Chicken Strips	87.00	254.52	7.39				
Regular Menu	6 piece Chicken McNuggets	96.00	456.00	6.25				
McCafe Menu	Moroccan Mint Green Tea (R)	330.00	144.00	381.77				
McCafe Menu	Moroccan Mint Green Tea (L)	456.00	112.00	281.44				
McCafe Menu	Moroccan Mint Green Tea (S)	279.00	145.00	411.09				
Regular Menu	9 piece Chicken McNuggets	144.00	201.50	125.25				
Breakfast Menu	Sausage Mc Muffin	112.00	236.50	147.72				
Regular Menu	5 piece Chicken Strips	145.00	266.50	166.99				
Breakfast Menu	Cappuccino (S)	201.50	297.50	183.61				
McCafe Menu	Latte (S)	236.50	307.50	190.03				
McCafe Menu	Flat White (S)	266.50	347.50	215.72				
Breakfast Menu	Cappuccino (R)	297.50	355.00	219.36				
McCafe Menu	Latte (R)	307.50						
McCafe Menu	Flat White (R)	347.50						
McCafe Menu	Cappuccino (L)	355.00						

Insert Data Cleaning

Key Steps

- Remove textual units (g/ml) $\rightarrow$ numeric conversion
- Handle missing values $\rightarrow$ mean imputation for numeric fields
- Standardize serving sizes and validate energy calculations

Quality Controls

- Range checks, duplicate detection, unit consistency
- Cross-validation with product spec sheets where available



Business Insight: Clean, standardized inputs are critical — downstream indices (NDI) and optimization depend on consistent units and validated nutrition fields.

Menu Pruning Using Z-Score Analysis

	A	B	C
1	Menu Category	Menu Items	Sodium Z-Score
2	Regular Menu	Ghee Rice with Mc Spicy Fried Chicken 1 pc	4.305993214
3	Gourmet Menu	Chunky Chipotle American Burger Chicken	3.263598486
4	Regular Menu	Chicken Maharaja Mac	3.154629116
5	Gourmet Menu	Chicken Cheese Lava Burger	2.922847295
6	Gourmet Menu	McSpicy Premium Chicken Burger	2.647760928
7	Regular Menu	Veg Maharaja Mac	2.466722989
8	Gourmet Menu	McSpicy Premium Veg Burger	2.292680562
9	Gourmet Menu	American Triple Cheese Chicken	2.185528758
10	Regular Menu	5 piece Chicken Strips	1.756245239
11	Gourmet Menu	American Triple Cheese Veg	1.716554689

Z-Score Analysis

Method: $Z = (x - \text{mean}) / \text{standard deviation}$ — applied to sodium, energy, and sodium-per-100g to flag extreme items.

Outlier Logic

Flag items with $|Z| > 2.5$ as candidates for review (nutritional risk or labeling error)

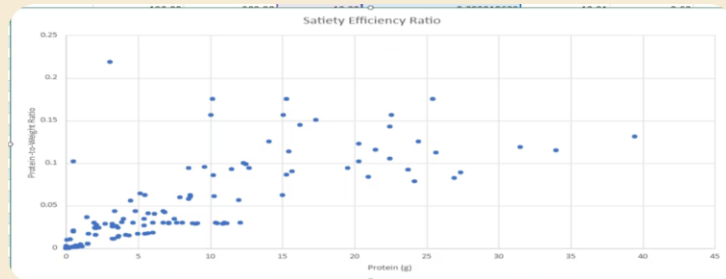
Use Case

Prioritize high-sodium and high-energy outliers for reformulation or removal.

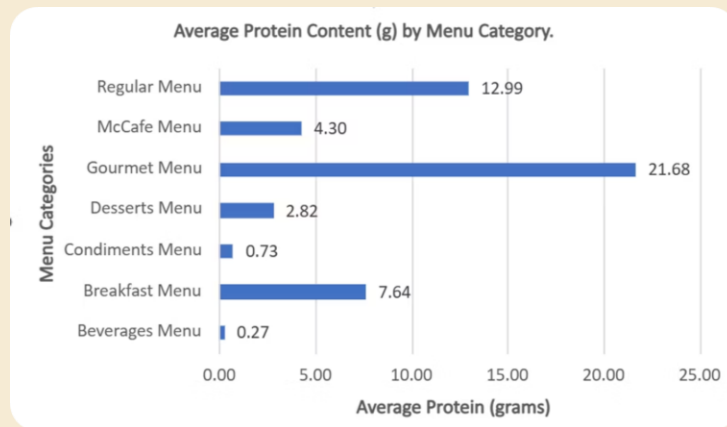


Managerial Recommendation: Use Z-score flags as a screen — combine with sales and margin data before making removal decisions.

Protein vs Satiety Efficiency Analysis



Protein vs Satiety Efficiency



Average Protein by Category Chart

Result: Correlation (Protein, Satiety Efficiency) = -0.0906



Business Insight: Weak negative correlation implies larger serving sizes do not reliably increase protein per calorie — opportunities exist to improve protein density without increasing energy.



Managerial Implication: Reformulate high-energy, low-protein items or re-bundle to improve perceived satiety while keeping calories controlled.

Nutri-Bundle Development

Bundle ID	Main Item	Side	Main Energy	Side Energy	Total Bundle Energy	Main Protein	Side Protein	Total Protein	Healthy Combo
Make Your Own Bundle	Green Chili Aliso Naan	Espresso Machiato	356.09	44.98	401.07	7.91	2.09	10	Standard
BD-001	FILET-O-FISH Burger	Coke Zero Can	348	0.99	348.99	15.44	0	15.44	Nutri-Bundle
BD-002	Mc chicken Burger	Black Coffee	400.8	5	405.8	15.66	0	15.66	Nutri-Bundle
BD-003	Egg McMuffin	Americano (S)	283	12	295	13.5	0	13.5	Nutri-Bundle
BD-004	Chicken Strips (3pc)	Small Coca-Cola	246	109	355	15.5	0	15.5	Nutri-Bundle
BD-005	6pc Chicken McNuggets	Regular Wedges	254	204	458	13.5	3	16.5	Nutri-Bundle

Healthy Bundle Optimization

Optimization Criteria

Total energy < 500 kcal;
Protein > 12 g; Sodium
within recommended
thresholds.

Design Approach

Combinatorial search
over menu items → filter
by nutritional thresholds
→ rank by satiety
efficiency and
commercial acceptability.

Commercial Considerations

Price sensitivity,
preparation complexity,
ingredient overlap for
supply chain efficiency.

Business Benefits: Nutri-Bundles create a measurable healthier offering that can attract health-conscious segments and support promotional campaigns without major menu overhaul.



Nutrient Density Index (NDI) & Health Tiering

Q3 $\text{=IF(P3 > 5, "Premium Health", IF(P3 > 3, "Balanced", "High-Risk"))}$

	Total carbohydrate (g)	Total Sugars (g)	Added Sugars (g)	Sodium (mg)	Sodium Z-Score	Nutrient-Density Index (NDI)	Strategic Health Tier
2	0	0	0	2.9	-0.759074671	#DIV/0!	
3	4	28.87	2.61	0.75	884.04	0.934032607	Premium Health
4	56	4.56	2.28	0	13.76	-0.71486815	Premium Health
5	3.3	3.3	1.65	0	17.19	-0.728873502	Premium Health
6	79	2.79	1.4	0	14.54	-0.734474138	Premium Health
7	11	12.7	0.58	0	873.89	1.081717236	Premium Health
8	43	0.72	0.54	0	178.95	-0.387002184	Premium Health
9	19	43.29	9.29	6.32	-0.765203669	5.780333784	Premium Health
10	0.1	2.68	0.29	0	477.22	0.243375894	Premium Health
11	79	0.28	1.4	0	13.84	-0.735935352	Premium Health
12	3.3	0.31	1.45	0	16.37	-0.73686579	Premium Health
13	46	10.5	0.33	0	313.25	-0.103166143	Premium Health
14	56	0.46	2.28	0	22.62	-0.71789748	Premium Health
15	15	4.02	0.44	0	715.83	0.747665676	Premium Health
16	99	15.74	0.48	0	469.87	0.227842053	Premium Health
17	3.3	3.3	1.65	0	17.69	-0.727816778	Premium Health
18	56	4.56	2.28	0	24.44	-0.713551005	Premium Health
19	79	2.79	1.4	0	14.95	-0.733607625	Premium Health
20	48	23.62	0.72	0	704.81	0.724375481	Premium Health
21	02	26.62	2.38	0.75	742.6	0.804242674	Premium Health
22	6.7	0.73	0.72	0	1155.05	1.756245239	Balanced
23	27	11.02	8.4	0	90.39	-0.574169122	Balanced
24	47	12.71	10.06	0	108.4	-0.536105927	Balanced
25	07	14.16	11.47	0	123.84	-0.503474293	Balanced

Conditional Formatting

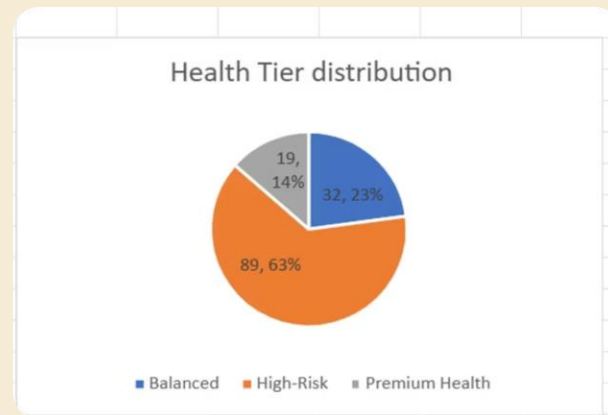
Manage Rules in selection +

As Cell Value contains High-Risk
Apply to: Q3

As Cell Value contains Balanced
Apply to: Q3

As Cell Value contains Pt...
Apply to: Q3

NDI Calculation



Health Tier Distribution Pie Chart

Formula: $\text{NDI} = (\text{Protein} / \text{Energy}) \times 100 - (\text{Sodium} / 1000)$

Health Tiers

Premium Health | Balanced | High-Risk — thresholds set on NDI quantiles and domain rules.

Use Cases

- Front-of-pack or digital menu labelling
- Tier-specific promotions and bundle suggestions

✔ **Strategic Insight:** NDI offers an operationally simple, interpretable metric to guide product development, customer communication, and regulatory reporting.

Conclusions & Strategic Recommendations

1. Menu Pruning

Remove or reformulate nutritionally inefficient outliers identified by Z-score screening — combine with profitability filters.

2. Launch Nutri-Bundles

Introduce 2–3 pilot bundles that meet energy & protein criteria; track uptake and margin impact.

3. Implement NDI-Based Labelling

Use NDI to create clear health tiers on menus and digital channels; support informed choices and brand trust.

4. Category Rebalancing

Prioritize product development toward high-NDI offerings in Gourmet/Regular categories where protein density is achievable.

Future Scope: integrate purchase behavior & POS data, develop ML recommendation engine, and operationalize real-time health scoring for dynamic menus.

Thank You — Questions & Answers